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A retrospective analysis of ketamine intravenous therapy for post-traumatic stress disorder in real-world care settings

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ABSTRACT

Objective: The study evaluated the effectiveness of Ketamine Intravenous Therapy (KIT) in reducing post-traumatic stress disorder (PTSD) and depression symptoms, using electronic health record data from outpatient psychiatric patients with PTSD.

Method: We analyzed data from 8,136 patients diagnosed with PTSD, 87 % (N = 7,070) of whom also had comorbid depression. Symptom severity was measured using the PTSD Checklist for DSM-5 (PCL-5) and the Patient Health Questionnaire-9 (PHQ-9). Linear mixed effects models were used to estimate changes in symptoms over time, with number of treatments as the primary predictor. Linear, quadratic, and logarithmic terms for treatment number were tested to explore different possible patterns of symptom change over time.

Results: The number of KIT treatments was associated with significant reductions in PTSD and depression symptoms. The model using a logarithmic transformation of treatment number provided the best fit to the data, suggesting diminishing improvements with additional treatments. By the 6th treatment, model-estimated mean reductions reached 22.4 points on the PCL-5 and 7.1 points on the PHQ-9, both exceeding thresholds for clinically meaningful change. The largest improvements occurred within the first three treatments, at doses of 0.6–0.8 mg/kg. While higher doses were often used in later treatments, our findings suggest that this did not appear to meaningfully enhance symptom reduction.

Conclusions: KIT is associated with reduced PTSD and depression symptoms in a community sample, with the most substantial gains occurring early in treatment. These findings support KIT as a promising treatment option for PTSD patients.

1. Introduction

Post-traumatic stress disorder (PTSD) remains a significant public health concern, with limited progress in the development of new treatments. Despite the availability of pharmacological and psychotherapeutic interventions, no new pharmacological treatments for PTSD have been approved since 2001, and existing treatments often yield suboptimal results. For example, although selective serotonin reuptake inhibitors (SSRIs) are commonly prescribed for PTSD, only approximately 60 % of patients respond to treatment, and only 20 %-30 % achieve complete remission following pharmacotherapy (Averill & Abdallah, 2022; Williams et al., 2022; Alexander, 2012; Zohar et al., 2002). This limited effectiveness often leads to polypharmacy in clinical practice (Krystal et al., 2017; Gasparyan et al., 2022). Additionally, while SSRIs do show moderate improvement in PTSD symptoms compared to placebo (Williams et al., 2022), their slow onset of action can increase suicide risk during the latency period before therapeutic benefits emerge (Averill & Abdallah, 2022). Alongside pharmacotherapy, psychotherapeutic approaches, particularly evidence-based trauma focused interventions such as prolonged exposure, cognitive processing therapy, and eye movement desensitization and reprocessing, have demonstrated clear efficacy for many patients (Lewis et al., 2020; Sciarrino et al.,

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2020). However, these treatments often take time to deliver and require active patient engagement, and some patients are unable to complete treatment due to elevated symptoms and functional impairments associated with PTSD (e.g., Lewis et al., 2020; Steenkamp, Litz, Hoge et al., 2015; Steenkamp, Litz, & Marmar, 2020; Stoycos et al., 2023; Storm & Christensen, 2021). Even among those who complete treatment, relapse and residual symptoms are common, underscoring the chronic and treatment-resistant nature of PTSD in many individuals (Kovacevic et al., 2023; Larsen et al., 2019). A systematic review of 51 randomized controlled trials found that 31 % of participants still reported clinical levels of PTSD symptoms post-treatment, and 59 % reported subthreshold levels (Larsen et al., 2019). Given these limitations, there has been growing interest in exploring mechanistically distinct interventions that target different neurobiological pathways.

Recent research has highlighted the complex neurobiological underpinnings of PTSD, involving dysregulation across multiple neurotransmitter systems, including glutamatergic, GABAergic, monoaminergic, and neuroendocrine pathways (Kelmendi et al., 2016; Prajapati et al., 2025; Friedman & Bernandy, 2017). This complexity suggests that targeting a single neurotransmitter system may be insufficient for effective treatment (Averill & Abdallah, 2022), and has prompted growing interest in novel and rapid-acting treatments. Among these, ketamine intravenous therapy (KIT) has emerged as a promising intervention, grounded in evidence that links PTSD to disruptions in glutamate signaling (Abdallah et al. 2019; Asim et al., 2021). KIT is a rapid, safe, and effective treatment (Walsh et al., 2021; Moore et al., 2022), initially used to treat depression, with growing evidence supporting its use across a range of psychiatric disorders (e.g., Derakhshanian et al., 2021; Murrough et al., 2013; Nikayin et al., 2022; Salloum et al., 2019). Early randomized controlled trials (RCTs) by Feder and colleagues (2014, 2021) demonstrated promising results in treating PTSD with use of KIT at 0.5 mg/kg. Beyond symptom reduction, KIT's rapid effect offers unique clinical value for populations with high functional impairment, acute distress, or elevated suicide risk, contexts where traditional pharmacotherapies may take weeks to become effective. (McInnes et al., 2022; Abbar et al., 2022; Borentain et al., 2022).

Comorbid major depressive disorder (MDD) is highly prevalent among patients with PTSD, with estimates ranging from around 40 % to over 60 % (Kostaras et al., 2017; Goetter et al., 2020). This comorbidity is associated with increased distress, social and economic costs, and suicide risk (e.g., Dold et al., 2017; Goetter et al., 2020, Flory & Yehuda, 2015; Pejuskovic et al., 2020) Thus, understanding how KIT affects both PTSD and depressive symptoms in PTSD patients is important. Additionally, examining how a comorbid MDD diagnosis might impact PTSD symptom patterns can provide valuable insight into treatment response.

Prior studies have begun to address these questions, though many remain limited by small sample sizes, single sites, or highly controlled settings (Borgogna et al., 2024; Almeida et al., 2024; Albuquerque et al., 2022). More recent naturalistic and retrospective studies have explored KIT's use in broader patient populations, describing dosing trends and real-world treatment trajectories (McInnes et al., 2022; Hietamies et al., 2023; Pfeiffer et al., 2024). These studies highlight the value of large-scale real-world data (RWD) to complement RCT findings, better reflecting how KIT is actually administered in community practice. Unlike RCTs, real-world clinical settings involve variable dosing schedules and flexible numbers of infusions, offering insight into how KIT is used in routine care. Real-world evidence (RWE) is essential for understanding treatment effectiveness outside controlled trials, as it captures more representative patient populations, treatment variations, and practical challenges such as adherence, cost, and accessibility (refer to McInnes, Marton, & Qian, 2024). RWD typically refers to information collected during routine clinical care, outside of controlled trials, using sources like electronic health records and insurance claims (Basch & Schrag, 2019). In practice, KIT typically follows an induction phase of 4-8 infusions over roughly 28 days, with 6 infusions being the most common (McInnes et al., 2022; Dai et al., 2022). This is often followed by maintenance infusions at longer intervals (Kryst et al., 2020; McInnes et al,. 2022; Phillips et al., 2019).

This study examines KIT's effects across diverse treatment patterns, providing a broader understanding of its real-world impact. We aimed to build on and extend the literature by exploring the real-world effectiveness of KIT in the treatment of PTSD using electronic health record (EHR) data drawn from a large national network of community clinics.

1.1. Study aims and objectives

The goal is to investigate KIT's effectiveness in reducing PTSD symptom severity in a real-world community outpatient sample and assess its impact on comorbid depression symptoms.

2. Methods

2.1. Patient sample & procedure

The current study analyzed de-identified data from a sample of patients who initiated KIT within Osmind's EHR platform, widely used by community interventional psychiatry practices. Patients and providers consented to the use of de-identified data for research purposes. This research sample included data from 325 community-based psychiatric practices across the United States that offered KIT. The clinics varied in size, patient demographics, and treatment approaches (e.g., dosing, frequency, survey administration), reflecting real-world care settings. States with the most clinics represented in the study were California, Texas, and Florida. The study protocol was deemed exempt from IRB review by the WCG Institutional Review Board under institutional guidelines and federal regulations (45 CFR 46.104).

The original cohort included patients aged 18 years or older as of the index date (date they began KIT treatment), who had a clinician-confirmed PTSD diagnosis (F43.1) and at least one documented KIT administration note in the EHR. Patients were asked to complete surveys during treatment including the PTSD Checklist for DSM-5 (PCL-5) and Patient Health Questionnaire (PHQ-9), sent electronically at various intervals of the clinicians' choosing. Two analytical samples were derived from the original cohort: The first sample (N = 1,340) included patients who completed a baseline PCL-5 and at least one follow-up PCL-5 within 15 days after an infusion. The second sample (N = 4,220) included patients who completed a baseline PHQ-9 and at least one follow-up PHQ-9 within 15 days after an infusion. A total of 1,306 patients completed both pre and post PCL-5 and PHQ-9 surveys. Fig. 1 presents a CONSORT diagram detailing the selection process for the two samples. For baseline surveys, we retained the survey completed within the month prior to the first treatment that was nearest in time to that treatment. Subsequent survey responses were included as follow-ups only if they occurred after a treatment and were within 15 days following the corresponding treatment. If multiple surveys fell within this window, we retained the one closest to the treatment date. Each treatment was linked to at most one follow-up survey. To quantify treatment exposure, we calculated the cumulative number of treatments received between baseline and each follow-up survey. This time-varying cumulative treatment count served as the primary predictor in our analysis, capturing total treatment exposure at the time each symptom assessment was completed.

2.2. Outcome measures

The primary outcome measure, the PCL-5 (Weathers et al., 2013), assesses PTSD symptom severity over the past month using 20 DSM-5-aligned items rated on a 5-point scale (0–4), with total scores ranging from 0 to 80. Scores above 30 often indicate a need for treatment. In our sample, the PCL-5 demonstrated excellent internal consistency ($\alpha = 0.96$).

The PHQ-9 (Kroenke et al., 2001) served as the secondary measure,

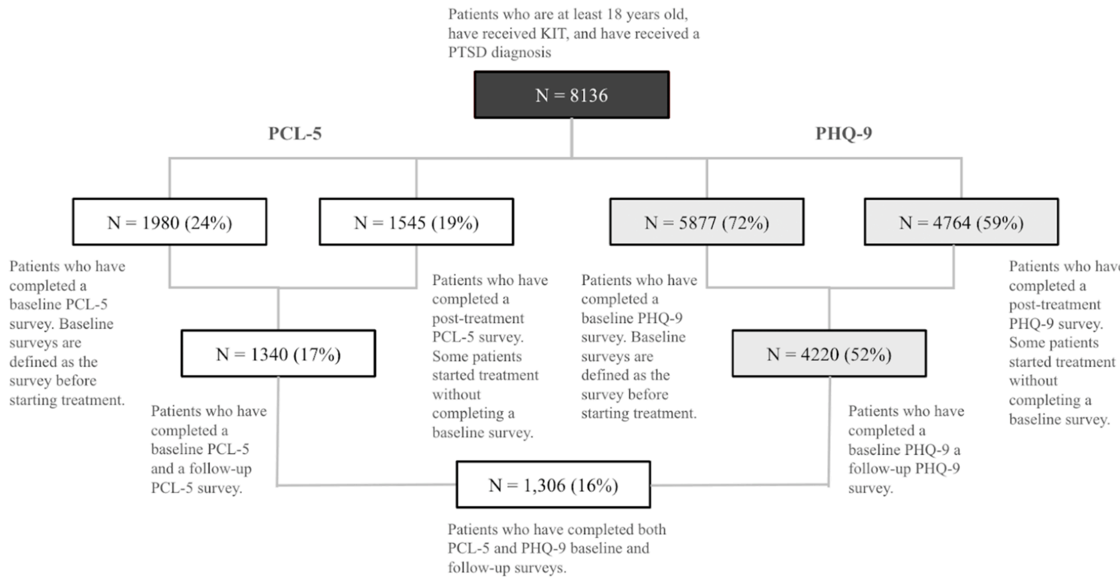

Fig. 1. CONSORT diagram mapping patient selection. All percentages are calculated using the initial sample (N = 8,136) as the denominator.

evaluating depressive symptom severity through nine items rated on a 4-point scale (0–3), with total scores ranging from 0 to 27. Higher scores indicate greater symptom severity. The PHQ-9 also showed strong internal consistency in our sample (α = 0.90).

2.3. Data analysis

Demographic and dosing characteristics were summarized using means and standard deviations for continuous variables, and proportions for categorical variables. KIT dosage was expressed in mg/kg and calculated using recorded body weight and dose administered per session. Clinical response was defined as a > 30 % reduction from baseline for the PCL-5 (Feder et al. 2014, 2021; Marx et al., 2022) and a > 50 % reduction for the PHQ-9 (Kroenke et al., 2001). In this study, treatment response rate refers to the proportion of patients who met these criteria at follow-up, in line with prior studies (e.g., Coley et al., 2020; Albott et al., 2018). Observed mean changes in PCL-5 and PHQ-9 scores were also calculated across treatment numbers (1st through 6th treatment) and stratified by the presence or absence of comorbid MDD (F32 or F33 ICD-10 diagnosis codes).

Two linear mixed effects models were fitted for PCL-5 and PHQ-9 scores separately using the 'lme4' package in R. Prior to modeling, two additional versions of the treatment number variable were created to test potential nonlinear relationships: a squared term (treatment number^2) and a log-transformed term (log(treatment number)). Three versions of the treatment number predictor were tested (i.e., linear, quadratic, and logarithmic), and the best fitting model was selected based on the lowest Bayesian Information Criterion (BIC). Models included fixed effects for treatment number, age (standardized, continuous), sex (binary: female/male), race (binary: Non-white/White), and BMI (standardized, continuous). To account for variability in treatment pacing and irregular intervals between infusions, we included treatment density (number of treatments divided by days since the index date) as a covariate and modeled its interaction with treatment number. This approach allowed us to capture differences in symptom trajectories related to treatment frequency, partially mitigating bias introduced by variable timing between treatments and symptom assessments. Random intercepts were specified to account for between-patient differences in baseline symptom severity, and random slopes were added to allow individual variation in symptom trajectories as a function of treatment number. Model results are presented as unstandardized fixed effect coefficients (b) with 95 % confidence intervals and p-values. Effect sizes are interpreted in terms of raw score point reductions, which can be interpreted relative to commonly used thresholds for clinically meaningful change (e.g., > 20 points on the PCL-5 and > 5 points on the PHQ-9; Salas et al., 2021; Kroenke, 2012).

To assess the robustness of the findings and whether PTSD symptom reductions were attributable specifically to PTSD rather than comorbid depression, sensitivity analyses were conducted on a restricted sample that excluded patients with a comorbid MDD diagnosis. Additionally, exploratory analyses examined the four PCL-5 symptom clusters to evaluate whether treatment effects were evenly distributed across PTSD symptom domains or if certain clusters were more responsive. All analyses were conducted using R/R studio (version 4.4.2).

2.4. Missing data

Survey completion rates for patients who were administered treatment and completed at least one PCL-5 were 100 % at baseline and treatment 1, 76 % for treatment 2, 61 % for treatment 3, 48 % for treatment 4, 41 % for treatment 5, 30 % for treatment 6. Completion rates for the PHQ-9 were 100 % at baseline and treatment 1, followed by 75 % for treatment 2, 54 % for treatment 3, 43 % for treatment 4, and 37 % for treatment 5, 29 % for treatment 6. We employed mixed-effects modeling, which utilizes maximum likelihood estimation to account for missing data related to unequal time points. This approach allows us to retain all available observations without artificially inflating the dataset (Baraldi and Enders, 2010; Little et al., 2016). To assess whether data are Missing at Random (MAR), we compared patients diagnosed with PTSD (our original cohort) who did not complete any PCL-5 surveys to those who did, assessing whether they differed significantly across demographic variables (e.g., age, sex, BMI) and clinical characteristics (e.g., baseline PCL-5, baseline PHQ-9). These comparisons revealed no significant differences between the groups ($p > 0.10$), suggesting the data are MAR.

3. Results

3.1. Descriptive statistics

The original cohort consisted of 8,136 patients diagnosed with PTSD. The sample was predominantly Caucasian (95 %) and female (66 %), with a mean age of 42.7 years ($SD = 12.8$). The most common comorbidities were major depressive disorders (MDD; 87 %), anxiety disorders

(85 %), and attention-deficit/hyperactivity disorder (ADHD, 32 %). Notably, a significant proportion of PTSD patients (87 %; N = 7,070 out of 8,136) had a comorbid diagnosis of MDD. Of the original cohort, N = 1,340 (16 %) patients completed a baseline and follow-up PCL-5 survey, and N = 4,220 (52 %) completed a baseline and follow-up PHQ-9 survey. These subsets form our analytical samples.

The majority of our original cohort (70.6 %) completed at least six treatments, which aligns with the common KIT practice model demonstrating six infusions as the most frequent (McInnes et al., 2022; Hietamies et al., 2023). A small, but notable, proportion (23.0 %) received at least 10 treatments, suggesting engagement in boosters or longer-term treatments. The mean dose administered during the first treatment is 0.64 mg/kg, which is similar to the 0.5 mg/kg used in the first clinical trial of KIT for depression and in later studies (Berman et al. 2000). As expected in real-world clinics, dosing tended to increase with additional treatments (Cusin et al., 2016). Subsequent dosages and the time intervals between treatments are summarized in Table 1.

We conducted an ANOVA focusing on the first 6 treatments to investigate whether dosage varied significantly across these early treatment sessions, as this period is most representative of the real-world standard induction phase (McInnes et al., 2022; Hietamines et al., 2023). Significant differences were found [$F(5, 37308) = 1612$; $p < 0.01$]. A post hoc Tukey's HSD test indicated that dosage significantly varied between all treatment session pairs ($p < 0.01$), suggesting systematic differences in dosing across the first six treatments.

3.2. Rates of treatment response

The proportion of patients achieving a clinical response in PTSD symptoms increased from 27.2 % of the sample after the first infusion to 77.3 % after the 6th infusion. Similarly, the proportion of patients achieving a clinical response in depression symptoms increased from 15.7 % after the first infusion to 60.5 % after the 6th infusion. Additional details on rates of treatment response are presented in Table 2.

3.3. Observed mean changes

An observed average reduction of 52.1 % in PCL-5 scores and 50.8 % in PHQ-9 scores from baseline was noted after six treatments. A detailed breakdown of these results are presented in Figs. 2 and 3 for PCL-5 and PHQ-9, respectively. Paired sample t-tests were conducted to assess significant differences between baseline scores and scores associated with the treatment number. All comparisons revealed significant differences, indicating that the mean scores for each treatment were significantly different from baseline scores for both the PCL-5 ($9.89 < t < 21.1$, $p < 0.001$) and the PHQ-9 ($16.1 < t < 37.3$, $p < 0.001$).

3.4. Model-estimated symptom changes

The mixed-effects model, with log transformed infusions, was identified as the best-fitting model for both the PCL-5 (BIC = 39,479) and the PHQ-9 (BIC = 91,921) compared to the linear model (PCL-5 BIC = 39,625; PHQ-9 BIC = 92,313) and the quadratic model (PCL-5 BIC = 39,590; PHQ-9 BIC = 92,153). Coefficients for the best-fitting models are reported in Table 3, and the model estimated scores are presented in Table 4, alongside the observed mean scores.

The model demonstrated a statistically significant reduction in PCL-5 scores as a function of the log-transformed number of treatments (b = -10.9, se = 0.43, $p < 0.001$), suggesting that each additional treatment leads to a decrease in PTSD symptoms, with the rate of improvement potentially slowing over time as more treatments are administered. The interaction between the number of treatments and density was significant (b = -2.71, se = 0.55, $p < 0.001$), such that patients with a higher density (more treatments within a smaller time window) demonstrated a

Table 1
Dosage information for the first 6 treatments.

Treatment number	N	Mean dosage (mg/kg)	Mean time between treatments (days)	Median time between treatments (days)
1	8136	0.64	–	–
2	7412	0.75	6.90	3.00
3	6979	0.84	7.14	4.00
4	6569	0.91	6.86	3.00
5	6165	0.97	8.15	4.00
6	5743	1.01	9.00	4.00

Table 2
Percentage of the sample achieving a clinical response for the PCL-5 and the PHQ-9. *Note:* Not all patients completed a survey after each treatment; as a result, the counts are not cumulative.

	Percentage achieved clinical response			
Treatment number	**PCL-5**		**PHQ-9**	
	Sample (N)	n (% Sample)	Sample (N)	n (% of Sample)
Baseline	1340	–	4220	–
1	681	185 (27.2 %)	1746	274 (15.7 %)
2	710	320 (45.1 %)	1862	564 (30.3 %)
3	688	394 (57.3 %)	1886	784 (41.6 %)
4	666	435 (65.3 %)	1818	895 (49.2 %)
5	716	516 (72.1 %)	1916	1108 (57.8 %)
6	436	337 (77.3 %)	1617	979 (60.5 %)

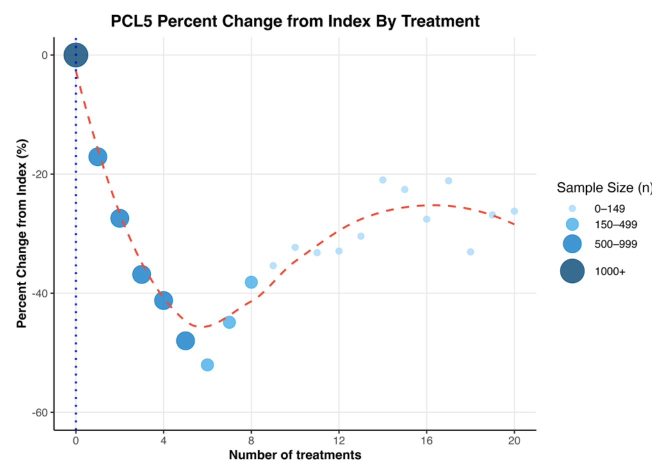

Fig. 2. Observed percent change in PCL-5 scores by the number of treatments.

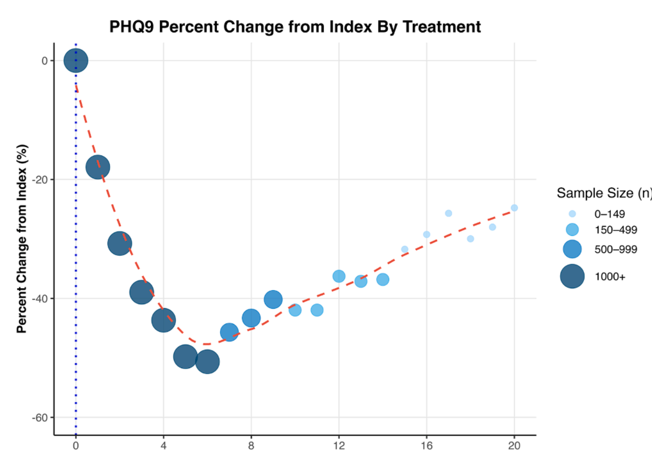

Fig. 3. Observed percent change in PHQ-9 scores by the number of treatments.

Table 3
Model predicting PCL-5 and PHQ-9 coefficients.

PCL-5				
Marginal R^2 / Conditional R^2	0.14/ 0.83			
Predictors	*Estimate*	*Std Error*	*df*	*t-value*
(Intercept)	45.9***	1.20	1760	38.3
Number of treatments (log)	-10.9***	0.43	1222	-25.1
Is non-white	0.38	1.03	1271	0.37
Is female	-0.84	0.50	1277	-1.68
Age (z)	-1.22*	0.49	1280	-2.50
BMI (z)	-0.68	0.44	1261	-1.54
Number of treatments * Treatment density	-2.71***	0.55	4364	-4.89

PHQ-9				
Marginal R^2 / Conditional R^2	0.10/ 0.78			
(Intercept)	15.4***	0.25	6091	61.7
Number of treatments (log)	-3.26***	0.09	3418	-36.7
Is non-white	-0.03	0.20	3985	-0.13
Is female	-0.15	0.10	4017	-1.48
Age (z)	-0.57***	0.09	3989	-6.20
BMI (z)	0.08	0.08	3873	0.99
Number of treatments * Treatment density	-1.57***	0.11	13720	-13.8

Note: **p < 0.01
* p < 0.05
*** p < 0.001

Table 4
Model estimated scores for PCL-5 and PHQ-9 by number of treatments.

	PCL-5		PHQ-9	
Treatment number	Mean observed scores	Model estimated scores	Mean observed scores	Model estimated scores
Baseline	46.6	45.9	16.4	15.4
1	38.6	37.9	13.4	12.8
2	33.8	33.3	11.3	11.4
3	29.4	30.0	9.98	10.3
4	27.4	27.4	9.21	9.52
5	24.2	25.3	8.21	8.86
6	22.3	23.5	8.07	8.30

steeper negative slope compared to patients with a lower density of treatment. Age was a significant covariate (b = -1.22, se = 0.49, $p < 0.05$), such that older age was associated with lower average symptom severity. Sex, BMI, and race were not significant. Model-estimated changes in PCL-5 closely aligned with the observed data for the first three treatments, estimating a 15.9 point reduction in PCL-5 scores from baseline (Mean = 45.9) to after the third infusion (Mean = 30.0), which was mostly consistent with the observed scores.

In a sensitivity analysis, we reran the model using only the sample without comorbid MDD. The results are consistent, with treatment number continuing to be a significant predictor of changes in PTSD symptoms (b = - 9.75, se = 1.42, $p < 0.001$). This suggests that the observed improvements are not solely driven by the treatment of co-occurring depression but also reflect a direct effect on PTSD symptoms. To further explore whether treatment effects were specific to PTSD rather than secondary improvements in depression, we also examined changes across the four core PTSD symptom clusters measured by the PCL-5: intrusion, avoidance, negative alterations in cognition and mood, and arousal and reactivity. The greatest reduction was observed in 'negative alterations in cognition and mood' (b = -4.27, $p < 0.001$), while the smallest was in avoidance symptoms (b = -1.12, $p < 0.001$). KIT was also associated with moderate improvements in intrusion (b = -2.40, $p < 0.001$), and arousal/reactivity symptoms (b = -2.78, $p < 0.001$).

The mixed-effects model for PHQ-9 scores, with log-transformed number of treatments, also showed a statistically significant reduction in depressive symptoms (b = -3.26, se = 0.09, $p < 0.001$). The interaction between the number of infusions and density was significant (b = -1.57, se = 0.11, $p < 0.001$). Age was also a significant covariate (b = -0.57, se = 0.09, $p < 0.001$), such that average PHQ-9 scores decreased by 0.57 points for each additional year of age. Sex, BMI, and race were not significant. Model-estimated changes in PHQ-9 closely aligned with the observed data, estimating a 5.1 point reduction in PHQ-9 scores from baseline (Mean = 15.4) to after the third infusion (Mean = 10.3), which was mostly consistent with the observed scores.

4. Discussion

The study explored the association between KIT treatment and changes in PTSD and depression symptoms in a large cohort of patients diagnosed with PTSD, most of whom also had comorbid depression. Findings suggest that KIT was associated with significant reductions in PTSD and depression symptoms, with the most pronounced improvements occurring within the first three treatments. These results align with earlier studies suggesting that KIT may help reduce PTSD symptom severity (Feder et al. 2014, 2021). Our study extends this literature by examining outcomes in routine clinical practice, where treatment protocols are less standardized and dosing tends to be higher than in RCTs. The improvements observed in our sample are encouraging, especially considering that 87 % of our sample had comorbid PTSD and MDD, a combination associated with greater symptom severity, chronic impairment, and higher dropout rates (Kessler, Chiu, Demler, & Walters, 2005; Post, Zoellner, Youngstrom, & Feen, 2011; Markowitz et al., 2015). Results from our sensitivity analyses, which excluded patients with comorbid MDD and separately modeled PTSD symptom clusters, suggest that KIT's effects on PTSD symptoms are not solely driven by improvements in depression.

The mixed effects models predicted clinically meaningful reductions in PTSD and depression symptoms associated with KIT treatments. Specifically, the PTSD model estimated a 15.9 point reduction from baseline in PCL-5 scores by the 3rd treatment and a 22.4 point reduction from baseline by the 6th treatment, exceeding the threshold for clinical significance (> 20 point reduction). Similarly, the depression model estimated a 5.1 point reduction in PHQ-9 scores after the 3rd treatment and a 7.1 point reduction after the 6th infusion, also meeting the threshold for clinically meaningful change (> 5 point reduction). The interaction between the number of treatments and density was a significant covariate in both models, such that patients with a higher density (more treatments within a smaller time window) demonstrated a steeper negative slope compared to patients with a lower density of treatment. Age was also a significant covariate in both models, suggesting that older age was associated with lower average symptom severity, consistent with previous findings (e.g., Reynolds et al., 2016). Age is sometimes discussed as a potential proxy for illness chronicity, and in some cases, KIT may be particularly effective in more chronic PTSD presentations (Du et al., 2022). Future studies should investigate whether chronicity or age influences KIT treatment outcomes.

The log-transformed pattern in the models suggests a non-linear relationship between the number of treatments and reductions in PCL-5 and PHQ-9 scores, with symptom improvements slowing down as more treatments are given. This is consistent with the broader phenomenon of diminishing returns often seen in repeated treatments. For example, Singh et al., (2016) found that four infusions produced similar antidepressant effects as six for the treatment of depression. However, it's also possible that early symptom reductions in our sample may partly reflect expectation or placebo effects (Aday et al., 2022), which can lessen with ongoing treatment. Nonetheless, the consistency and magnitude of observed symptom reductions in our large, highly comorbid sample suggest that the improvements may still reflect a meaningful treatment signal, though causal conclusions cannot be drawn.

The doses used in our sample were higher than those typically used in clinical trials (e.g., ~0.5 mg/kg; Feder et al., 2021). The relatively rapid escalation of dosing during treatment from an initial mean of 0.6 mg/kg to approximately 1 mg/kg among real-world clinics may help explain why some prior studies using lower doses (0.2-0.5 mg/kg) failed to demonstrate an effect of KIT for PTSD (Abdallah et al. 2022). We observed the most substantial improvements in both PTSD and depression symptoms during the first three infusions, when mean doses ranged from 0.6 to 0.8 mg/kg. While this study did not directly model the relationship between dose and symptom reduction, dose escalation commonly occurs alongside additional treatments (Cusin et al., 2016). Yet, our findings suggest that beyond these early treatments, additional infusions may provide diminishing symptom improvement. Further research is needed to clarify the dose-response relationship and to determine whether the observed trend is driven by dose escalation, cumulative treatment effects, or other factors.

4.1. Limitations

Several limitations should be considered. First, as a retrospective, observational study without a control group, we cannot fully account for potential confounders, such as concurrent treatments, spontaneous remission, or placebo effects. Early symptom improvements may partly reflect expectancy or placebo effects rather than a direct effect of ketamine treatment. Our assessments showed improvements after the first treatment, which may be partly driven by patient expectations. Future studies could better disentangle these effects by including structured expectancy assessments or using quasi-experimental designs (e.g., active comparators or delayed treatment). Nonetheless, expectancy effects are inherent to real-world care and contribute to the overall therapeutic response. Our design enhances ecological validity by capturing real-world patient experiences (e.g., variations in dosing, measurement-based care), complementing findings from more controlled trials (e.g., McInnes, Marton, and Qian, 2024). Another limitation is that our sample was predominantly Caucasian and female, which may limit generalizability to more diverse populations. Additionally, only 16.4 % of PTSD patients receiving KIT in our network completed both pre-and post--treatment PCL-5 assessments, though patient characteristics were similar between those who did and did not complete both assessments. Additionally, while patients were identified by PTSD diagnosis codes, we could not confirm primary diagnoses or which symptoms clinicians prioritized. Some patients may have also been receiving concurrent behavioral therapies, potentially influencing outcomes.

We also acknowledge that while the interaction between the number of treatments and density was a significant covariate in our models, time was not our primary focus. Future research should more directly explore how temporal dimensions influence treatment response. Another limitation is the considerable variability in patient adherence to the treatment protocol. This heterogeneity in treatment delivery and timing introduces potential confounding factors and increases the variance in clinical outcomes, which may impact the generalizability and interpretability of the results. While including treatment density in the models helped account for variability in treatment pacing, it does not fully eliminate potential bias introduced by irregular survey completion and the simplified approach of linking a single survey response to each treatment. Additionally, as our analyses focused on treatment episodes where follow-up surveys could be temporally paired with a recent treatment, cases where patients completed surveys without a recent treatment were excluded. This limits our ability to assess the durability of treatment gains once treatment stopped. Future research will be important to better evaluate maintenance and longer-term outcomes. Overall, our study leverages one of the largest longitudinal PTSD cohorts treated with KIT, offering real-world insights that strengthen the generalizability of our findings. While these data add to the growing evidence base for KIT in PTSD, these limitations can be best addressed through a well-powered randomized controlled trial.

5. Conclusion

Our findings carry important clinical implications. In our sample, a substantial proportion of patients (77.3 % for PCL-5 and 60.5 % for PHQ-9) showed a clinical response within the first six treatments, highlighting the potential for KIT to provide fast-acting relief, even among patients with comorbid PTSD and MDD (87 % of our sample). Notably, reductions in the PCL-5 were similar in patients without comorbid depression, suggesting a direct effect of KIT on PTSD symptoms rather than a secondary effect of treating depression. Given the critical unmet need in PTSD treatment, the results from this study adds to evidence suggesting KIT may be a promising alternative, offering a much-needed solution for individuals battling this condition.

CRediT authorship contribution statement

Lynne Alison McInnes: Writing – original draft, Supervision, Methodology, Conceptualization. **Robert Mark Berman:** Writing – review & editing, Methodology. **Matthew Worley:** Writing – review & editing, Methodology, Formal analysis. **Emily Shih:** Writing – original draft, Visualization, Methodology, Formal analysis.

Declaration of competing interest

Dr. McInnes, Dr. Worley, and Dr. Shih were paid employees of Osmind during the study. Dr. Berman is a shareholder and advisor for MindImmune Therapeutics, Gilgamesh Pharmaceuticals, Osmol Therapeutics, Biohaven Pharmaceuticals, Freedom Biosciences, and Gaba Therapeutics. All authors hold equity in Osmind. The study was fully funded by Osmind.

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